

EC400

Lecture 3

# Statistics II

Probability and Statistical Inference ■ Tom Glinnan

## Plan for today

- Common distributions and their economic applications.
- Order statistics: sample minima and maxima. non-examinable
- Covariance matrices, normal vectors,  $\chi^2$ ,  $t$  and  $F$ .
- Finite moments, heavy tails and Jensen's inequality.
- Regression notation, OLS and error covariance matrices.

### Appendix non-examinable

Deriving OLS by minimising squared residuals.

# Distributions for modelling

## Why are we looking at these distributions?

- These 12 distributions describe choices, counts, waiting times and economic uncertainty.
- In **economic models**, including macro models, they capture shocks and differences between people or firms.
- In **maximum likelihood estimation (MLE)**, they help us find parameters that make the data most plausible.
- There are many more: choose a distribution to suit the economic question.

**Aim for intuition, not memorisation.**

Memorising the PDFs, PMFs and other formulas in this gallery is **non-examinable**.

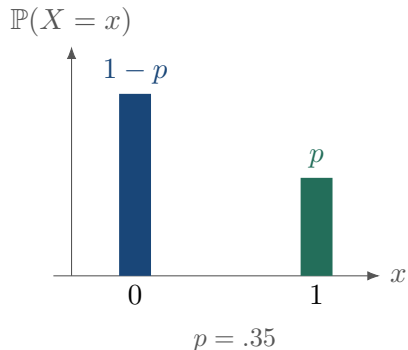
## The Bernoulli distribution

- A Bernoulli variable records **one binary outcome**:

$$X \sim \text{Bernoulli}(p),$$

$$X \in \{0, 1\}.$$

- $\mathbb{P}(X = 1) = p$  and  $\mathbb{P}(X = 0) = 1 - p$ .
- $\mathbb{E}[X] = p$  and  $\text{Var}(X) = p(1 - p)$ .



Whether someone works is a yes/no outcome.

- In Mroz's study of married women, code working as 1 and not working as 0.
- Probit regression links the chance of working to education, children and income.
- Maximum likelihood fits those probabilities to the observed choices.

Example: Mroz (1987), The Sensitivity of an Empirical Model of Married Women's Hours of Work to Economic and Statistical Assumptions.

## The binomial distribution

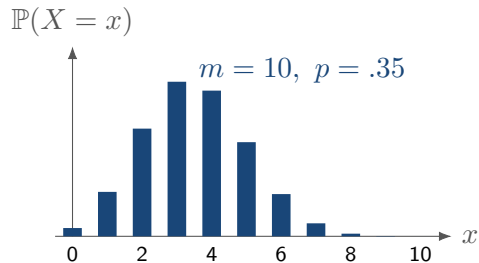
- Count successes across  $m$  independent Bernoulli trials with the **same success probability**  $p$ :

$$X \sim \text{Binomial}(m, p).$$

- For  $x = 0, \dots, m$ ,

$$\mathbb{P}(X = x) = \binom{m}{x} p^x (1 - p)^{m-x}.$$

$$\mathbb{E}[X] = mp, \quad \text{Var}(X) = mp(1 - p).$$



## Binomial: option pricing

An option pays according to a future share price.

- Model the share price as moving up or down at each step.
- The binomial counts up moves over a fixed number of steps.
- Use the resulting payoffs to price the option. Pricing probabilities need not be real-world forecasts.

Example: Cox, Ross and Rubinstein (1979), Option Pricing: A Simplified Approach.

# The Poisson distribution

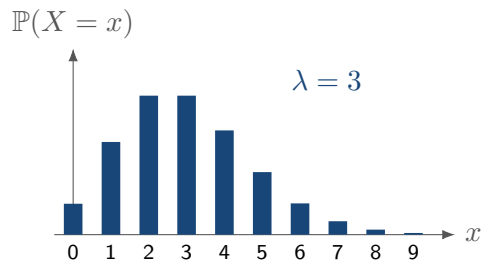
- Count events arriving in a fixed window:

$$X \sim \text{Poisson}(\lambda).$$

- For  $x = 0, 1, \dots$ ,

$$\mathbb{P}(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}.$$

- $\mathbb{E}[X] = \text{Var}(X) = \lambda.$
- Approximates a binomial with many trials and rare successes, keeping  $mp = \lambda.$



Do firms that spend more on research produce more patents?

- Patents are counts:  $0, 1, 2, \dots$
- Poisson regression links expected patents to R&D and firm characteristics.
- Fit by maximum likelihood; predicted means stay positive and zero counts fit naturally.

Example: Hausman, Hall and Griliches (1984), Econometric Models for Count Data with an Application to the Patents–R&D Relationship.

## The negative binomial distribution

- Count trials up to the  $r$ th success in independent Bernoulli trials ( $r = 1, 2, \dots$ ):

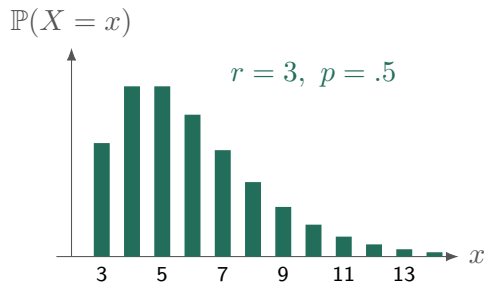
$$X \sim \text{NegBin}(r, p).$$

- For  $x = r, r + 1, \dots$  and  $0 < p < 1$ ,

$$\mathbb{P}(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}.$$

- When  $r = 1$ , we wait only for the first success: the **geometric** case.

$$\mathbb{E}[X] = \frac{r}{p}, \quad \text{Var}(X) = \frac{r(1-p)}{p^2}.$$



## Negative binomial: patent counts

Similar research budgets can produce very different patent counts.

- The full Poisson model ties the conditional variance to the mean.
- Negative binomial regression allows extra spread around expected patents.
- It can capture unobserved differences in how successfully firms turn research into inventions.

Count version:  $Y = X - r = 0, 1, \dots$ ;  $X$  on the previous slide counts total trials.

Example: Hausman, Hall and Griliches (1984), Econometric Models for Count Data with an Application to the Patents–R&D Relationship.

## The geometric distribution

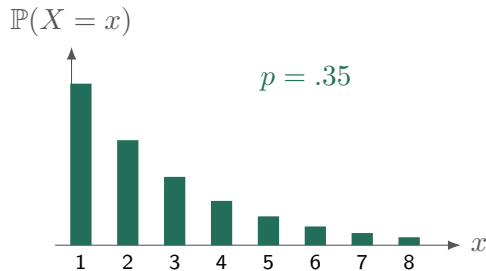
- The  $r = 1$  negative binomial case: trials up to the **first success**:

$$X \sim \text{Geometric}(p).$$

- For  $x = 1, 2, \dots$ ,

$$\mathbb{P}(X = x) = (1 - p)^{x-1}p.$$

- $\mathbb{E}[X] = 1/p$ ; each unsuccessful trial restarts the same waiting problem.



## Geometric: job-finding

How many months until an unemployed worker finds a job?

- Give the worker the same chance  $p$  of finding a job each month.
- With independent monthly opportunities, the first successful month is geometric.
- For illustration,  $p = 1/4$  gives an average wait of four months.

Discrete-time illustration based on Shimer (2012), Reassessing the Ins and Outs of Unemployment.

# The exponential distribution

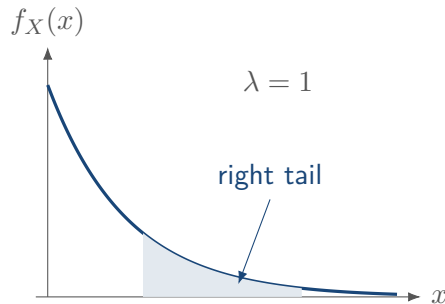
- A positive waiting time with rate  $\lambda$ :

$$X \sim \text{Exponential}(\lambda).$$

Its density is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

- $\mathbb{E}[X] = 1/\lambda$  and  $\text{Var}(X) = 1/\lambda^2$ .
- It is **memoryless**: elapsed waiting time leaves the remaining distribution unchanged.



## Exponential: job-finding

The same job-finding question, now measured in continuous time.

- Jobs can arrive at any time, at a steady rate.
- Geometric counts months; exponential measures the exact wait, with the **rate held fixed**.
- A higher job-finding rate means a shorter average wait. Changing rates can change this distribution.

Example: Shimer (2012), Reassessing the Ins and Outs of Unemployment.

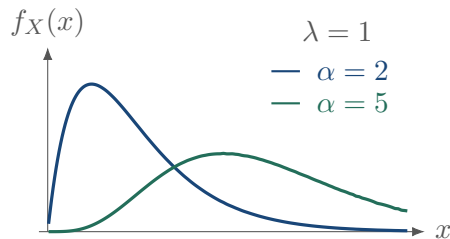
# The gamma distribution

- A flexible positive distribution with shape  $\alpha$  and rate  $\lambda$ :

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}.$$

Its support is  $x > 0$ .

- $\mathbb{E}[X] = \alpha/\lambda$  and  $\text{Var}(X) = \alpha/\lambda^2$ .
- For integer  $\alpha$ , add  $\alpha$  independent exponential waits with rate  $\lambda$ : time to the  $\alpha$ th Poisson arrival.



## Gamma: health spending

Many medical bills are modest; a few are very large.

- Manning and Mullahy compare methods for modelling positive health spending.
- Gamma regression captures positive, skewed amounts.
- A log link keeps predicted means positive. Whether someone spends anything at all is a separate question.

Example: Manning and Mullahy (2001), Estimating Log Models: To Transform or Not to Transform?.

## The uniform distribution

- $X$  is **uniform** on  $[a, b]$  if every interval of the same length is equally likely.

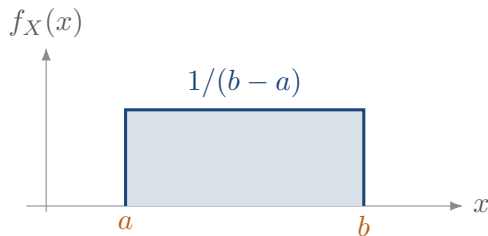
$$X \sim U[a, b].$$

- Its density is flat:

$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

- The area under the rectangle is one.

$$\mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$



## Uniform: customer locations

Two shops compete for customers along a street.

- Assume customers are evenly spread along the street.
- A stretch covering 20% of its length contains 20% of customers.
- Prices and travel costs determine who buys where; market shares are easy to calculate.

Example: Hotelling (1929), Stability in Competition.

## The beta distribution

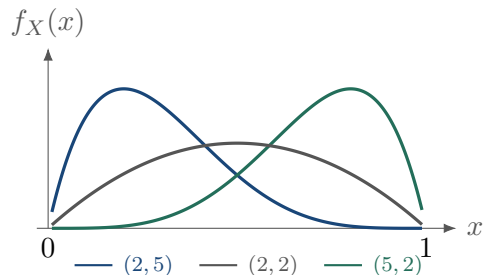
- A distribution on the **unit interval**:

$$X \sim \text{Beta}(a, b), \quad a, b > 0.$$

- Its density for  $0 < x < 1$  is

$$f_X(x) = \frac{x^{a-1}(1-x)^{b-1}}{B(a, b)}.$$

- $\mathbb{E}[X] = a/(a + b)$ .
- With  $a = b = 1$ , this is **uniform** on  $[0, 1]$ ; other values change the shape.



## Beta: sticky prices

How likely is a firm to keep its price unchanged this quarter?

- The keep/change decision is **Bernoulli**; its probability is unknown.
- In the Smets–Wouters model, a **beta prior** describes beliefs about that probability before seeing data.
- The distribution keeps the probability between zero and one; the data then update those beliefs.

Example: Smets and Wouters (2007), Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach.

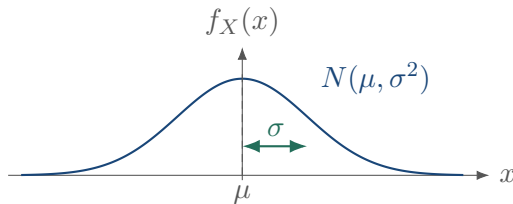
# The normal distribution

- Write

$$X \sim N(\mu, \sigma^2).$$

- $\mu$  controls the centre.
- $\sigma^2$  controls the spread.
- The standard normal is

$$Z \sim N(0, 1).$$



## Normal: macro shocks

A macro model needs unexpected shocks to generate fluctuations.

- Model fresh surprises to technology or spending as normal.
- Small surprises are common; large surprises are rare; **either sign is possible**.
- Feed these shocks through the model to simulate output, investment and inflation.

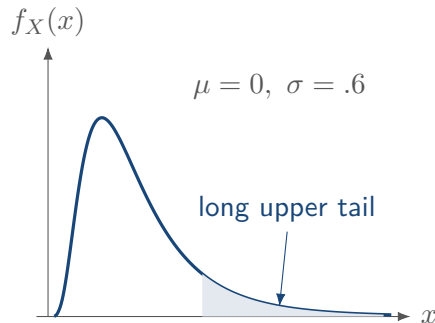
Example: Smets and Wouters (2007), Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach.

# The log-normal distribution

- A positive variable is log-normal when its logarithm is normal:

$$\log X \sim N(\mu, \sigma^2).$$

- Its density is **right-skewed**, with a long upper tail.
- Median =  $e^\mu$  and mean =  $e^{\mu+\sigma^2/2}$ .



## Log-normal: share prices

Share prices build up through percentage changes.

- A 10% rise **multiplies** the price by 1.10.
- In the Black–Scholes model, accumulated log changes are normal, so future prices are log-normal.
- Return to option pricing: the binomial tree gives discrete prices; this model gives a continuous range.

Example: Black and Scholes (1973), The Pricing of Options and Corporate Liabilities.

## The Pareto distribution

- A heavy-tailed distribution with minimum  $x_m > 0$  and shape  $\alpha > 0$ :

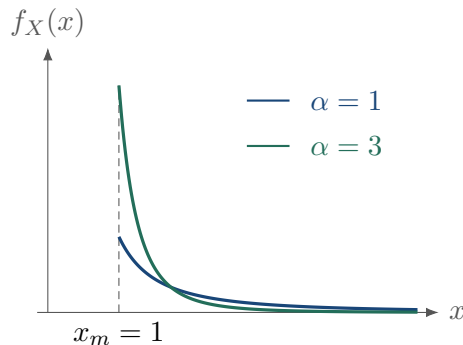
$$f_X(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, \quad x \geq x_m.$$

- Its upper-tail probability follows a power law:

$$\mathbb{P}(X > x) = \left(\frac{x_m}{x}\right)^\alpha, \quad x \geq x_m.$$

- Smaller  $\alpha$  means a **heavier upper tail**.

$$\mathbb{E}[X] = \frac{\alpha x_m}{\alpha - 1} \quad (\alpha > 1); \quad \text{infinite otherwise.}$$



## Pareto: wealth inequality

How much wealth is held by the very richest households?

- Household surveys can miss the very wealthy, understating wealth at the top.
- Fit a **Pareto tail** using survey data and rich lists to estimate the missing wealth at the top.
- This helps estimate wealth concentration, such as the share owned by the **top 1%**.

Example: Vermeulen (2018), How Fat Is the Top Tail of the Wealth Distribution?.

## Counts and waiting times

| Distribution      | What it captures                | Connection                                                                    |
|-------------------|---------------------------------|-------------------------------------------------------------------------------|
| Bernoulli         | One yes/no outcome              | Building block of a binomial count.                                           |
| Binomial          | Successes in $m$ trials         | Sum of $m$ iid Bernoulli variables.                                           |
| Poisson           | Events in a fixed window        | Binomial limit: $m \rightarrow \infty$ , $p \rightarrow 0$ , $mp = \lambda$ . |
| Negative binomial | Trials until the $r$ th success | Sum of $r$ iid geometric waits.                                               |
| Geometric         | Trials until the first success  | Negative binomial with $r = 1$ .                                              |
| Exponential       | Continuous time to an event     | Wait to the next Poisson arrival.                                             |
| Gamma             | Accumulated positive waits      | Sum of $\alpha$ iid exponentials (integer $\alpha$ ).                         |

Count events in a window, or measure the wait until they occur.

## Bounds, shocks and tails

| Distribution | What it captures                                             |
|--------------|--------------------------------------------------------------|
| Uniform      | Equal probability in equal-length intervals.                 |
| Beta         | Proportions or uncertain probabilities on $(0, 1)$ .         |
| Normal       | Symmetric, bell-shaped uncertainty.                          |
| Log-normal   | Positive outcomes built from multiplicative changes.         |
| Pareto       | A power-law upper tail with economically important extremes. |

Uniform is a special beta

$$\text{Beta}(1, 1) = U[0, 1].$$

Exponentiate a normal

$$Z \sim N(\mu, \sigma^2) \Rightarrow e^Z \text{ is log-normal.}$$

The Pareto tail decays more slowly than the log-normal tail.

# Order statistics

non-examinable

Given a sample  $X_1, \dots, X_n$ , arrange its values from lowest to highest:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}.$$

These are the **order statistics**; in particular,

$$L = X_{(1)} = \min_i X_i, \quad H = X_{(n)} = \max_i X_i.$$

If the data are iid, this gives us two distinct facts:

- **Identically distributed:**  $F_{X_1}(x) = \dots = F_{X_n}(x) = F(x)$ .
- **Independent:** probabilities involving all observations can be multiplied.

Order statistics appear in auctions, rankings, reliability and extreme outcomes.

## Sample extremes non-examinable

For the **maximum**  $H = X_{(n)}$ ,

$$\begin{aligned} F_H(x) &= \mathbb{P}(X_1 \leq x, \dots, X_n \leq x) \\ &\stackrel{\text{independence}}{=} \prod_{i=1}^n \mathbb{P}(X_i \leq x) \stackrel{\text{same distribution}}{=} F(x)^n. \end{aligned}$$

For the **minimum**  $L = X_{(1)}$ , it is easiest to use the complement:

$$\begin{aligned} 1 - F_L(x) &= \mathbb{P}(X_1 > x, \dots, X_n > x) \\ &\stackrel{\text{independence}}{=} \prod_{i=1}^n \mathbb{P}(X_i > x) \stackrel{\text{same distribution}}{=} [1 - F(x)]^n. \end{aligned}$$

Hence

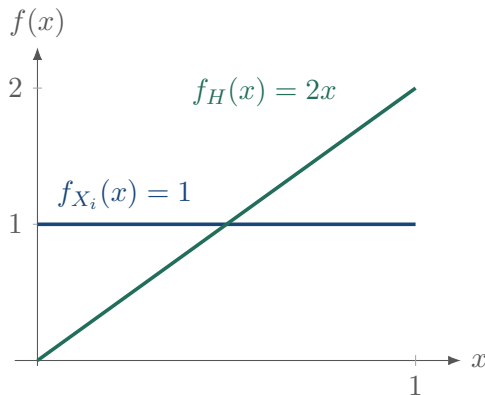
$$F_H(x) = F(x)^n, \quad F_L(x) = 1 - [1 - F(x)]^n.$$

## The maximum of two uniforms non-examinable

Let  $X_1, X_2 \stackrel{\text{iid}}{\sim} U[0, 1]$  and  $H = \max\{X_1, X_2\}$ . For  $0 \leq x \leq 1$ ,

$$\begin{aligned} F_H(x) &= \mathbb{P}(X_1 \leq x, X_2 \leq x) \\ &\stackrel{\text{independence}}{=} \mathbb{P}(X_1 \leq x) \mathbb{P}(X_2 \leq x) \\ &\stackrel{\text{both uniform}}{=} x \times x = x^2. \end{aligned}$$

Thus  $f_H(x) = 2x$ : the maximum is more likely to be near one.



**Auction example:** with two iid bidder valuations,  $H$  is the highest valuation.

# Distributions for testing

## Why these distributions?

These distributions often describe statistics used in hypothesis tests.

- We will usually define them by **combining random variables**, often normals: taking sums, squares or ratios.
- They have PDFs, but the construction is the useful starting point here.
- For now, treat these as **definitions** and build intuition for how the pieces fit together.

In econometrics and other courses, you will see why these combinations arise and how they are used in testing.

## Covariance as matrix products

Covariance measures how two variables move together.

$$\text{Cov}(u_1, u_2) = \mathbb{E}[(u_1 - \mathbb{E}[u_1])(u_2 - \mathbb{E}[u_2])].$$

Centre each variable:  $d_i = u_i - \mathbb{E}[u_i]$ . Collect their products:

$$\underbrace{\begin{pmatrix} d_1 \\ d_2 \end{pmatrix}}_d \underbrace{\begin{pmatrix} d_1 & d_2 \end{pmatrix}}_{d'} = \begin{pmatrix} d_1^2 & d_1 d_2 \\ d_2 d_1 & d_2^2 \end{pmatrix}.$$

## The covariance matrix

For  $u = (u_1, \dots, u_n)'$  with finite second moments, let  $\mu = \mathbb{E}[u]$  and  $d = u - \mu$ .

$$\text{Var}(u) = \mathbb{E}[dd'] = \mathbb{E}[(u - \mu)(u - \mu)'].$$

Each entry is a covariance:

$$\omega_{ij} = \mathbb{E}[d_i d_j] = \text{Cov}(u_i, u_j).$$

$$\text{Var}(u) = \begin{pmatrix} \omega_{11} & \omega_{12} & \cdots & \omega_{1n} \\ \omega_{21} & \omega_{22} & \cdots & \omega_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{n1} & \omega_{n2} & \cdots & \omega_{nn} \end{pmatrix}.$$

- **Diagonal:**  $\omega_{ii} = \text{Var}(u_i)$ .
- **Off diagonal:** covariance between different variables.

The matrix is **symmetric**:  $\omega_{ij} = \omega_{ji}$  because  $d_i d_j = d_j d_i$ .

## Normal vectors

A vector of random variables is a **random vector**.

$$X = \begin{pmatrix} X_1 \\ \vdots \\ X_k \end{pmatrix}$$

A multivariate normal is written  $X \sim N_k(\mu, \Sigma)$ :

- $\mu = \mathbb{E}[X]$ : the  $k \times 1$  mean vector.
- $\Sigma = \text{Var}(X)$ : the  $k \times k$  covariance matrix.

For a **fixed matrix**  $A$  of compatible dimensions,

$$\mathbb{E}[AX] = A\mathbb{E}[X], \quad \text{Var}(AX) = A \text{Var}(X) A'.$$

These rules hold for **any** random vector (not just normals).

## Normal vectors: useful properties

- Marginals of a multivariate normal are normal:

$$X \sim N_k(\mu, \Sigma) \quad \Rightarrow \quad X_j \sim N(\mu_j, \Sigma_{jj}).$$

- Linear transformations stay normal:

$$Y = c + BX \quad \Rightarrow \quad Y \sim N(c + B\mu, B\Sigma B').$$

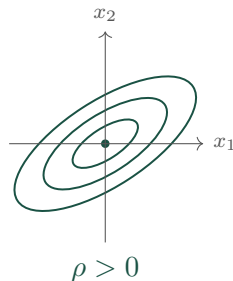
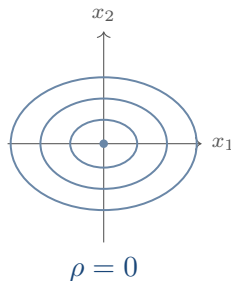
- Conditional distributions are normal too.
- Special warning: for jointly normal variables,

$$\text{Cov}(X_1, X_2) = 0 \quad \Rightarrow \quad X_1 \perp\!\!\!\perp X_2.$$

This implication is special to joint normality, not true in general.

## Covariance shapes the density

- In two dimensions, the density is a hill over the  $(x_1, x_2)$  plane.
- Horizontal slices of that hill are contours.
- Diagonal entries of  $\Sigma$  set spread in each direction.
- Off-diagonal entries tilt the contours.



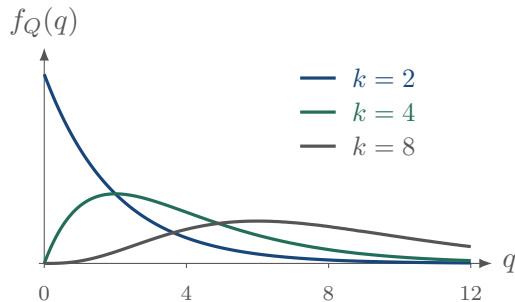
$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

## The $\chi^2$ distribution

For independent standard normals  $Z_1, \dots, Z_k$ ,

$$Q = \sum_{j=1}^k Z_j^2 \implies Q \sim \chi_k^2.$$

- $k$  is the **degrees of freedom**.
- It counts the squared independent standard normals.
- As  $k$  grows, the distribution shifts right and becomes less skewed.



## Why a $t$ distribution?

Under suitable assumptions, under  $H_0: \beta = 0$ ,

$$\hat{\beta} \sim N(0, V) \quad \Longrightarrow \quad Z = \frac{\hat{\beta}}{\sqrt{V}} \sim N(0, 1).$$

But  $V$  is unknown: we use an estimated variance  $\hat{V}$ .

$$T = \frac{\hat{\beta}}{\sqrt{\hat{V}}} = \frac{\hat{\beta}}{\sqrt{V}} \frac{\sqrt{V}}{\sqrt{\hat{V}}} = \frac{Z}{\sqrt{\hat{V}/V}}.$$

If  $Q = k\hat{V}/V \sim \chi_k^2$  and  $Q \perp\!\!\!\perp Z$ , then

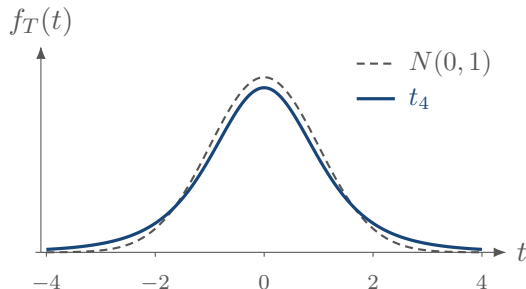
$$T = \frac{Z}{\sqrt{Q/k}} \sim t_k.$$

## Student's $t$ distribution

For  $Z \sim N(0, 1)$  and  $Q \sim \chi_k^2$  with  $Z \perp\!\!\!\perp Q$ ,

$$T = \frac{Z}{\sqrt{Q/k}} \implies T \sim t_k.$$

- Numerator: normal noise.
- Denominator: estimated scale.
- Small  $k$  means more uncertainty in the scale, so the **tails are heavier**.

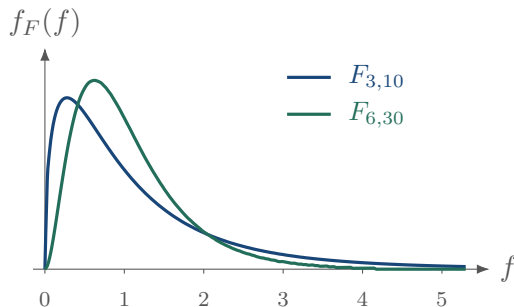


## The $F$ distribution

For independent  $Q_1 \sim \chi_{d_1}^2$  and  $Q_2 \sim \chi_{d_2}^2$ ,

$$F = \frac{Q_1/d_1}{Q_2/d_2} \implies F \sim F_{d_1, d_2}.$$

- A ratio of two variance-like quantities.
- Nonnegative and typically **right-skewed**.
- Appears in variance comparisons and joint tests.



## Four connected distributions

- Start from standard normal random variables.

$$Z \sim N(0, 1).$$

- Squared and summed normals give chi-squared variables:

$$Q = \sum_{j=1}^{\nu} Z_j^2 \sim \chi_{\nu}^2.$$

- A normal divided by an estimated scale gives a  $t$  variable:

$$T = \frac{Z}{\sqrt{Q/\nu}} \sim t_{\nu}.$$

- Ratios of scaled chi-squared variables give  $F$  variables:

$$F = \frac{Q_1/d_1}{Q_2/d_2} \sim F_{d_1, d_2}, \quad T_{\nu}^2 \sim F_{1, \nu}.$$

# Existence of Moments

## Why do we need finite moments?

The LLN needs finite moments; we use it to prove **consistency**.

What does it mean for a moment to be finite?

For  $X \sim t_k$ :

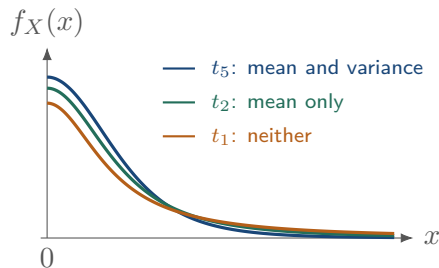
- A finite mean requires  $k > 1$ :

$$\mathbb{E}[|X|] < \infty.$$

- A finite variance requires  $k > 2$ :

$$\mathbb{E}[X^2] < \infty.$$

Similar-looking densities can have very different **tails**.

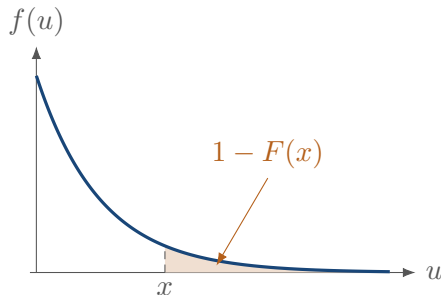


## Vanishing tail area

A distribution has total probability one:

$$P(X \leq x) \rightarrow 1 \quad \Leftrightarrow \quad \int_x^\infty f(u) du \rightarrow 0 \quad \text{as } x \rightarrow \infty.$$

- For the decreasing tails we consider, this requires  $f(x) \rightarrow 0$ .
- If the density stayed above a positive level, its tail would have **infinite area**.
- But how **quickly** does it approach zero?



## Second moments: two forces

A finite second moment means that this **weighted area** is finite:

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx < \infty.$$

Larger outcomes get more weight

$$x^2 \longrightarrow \infty.$$

Squaring makes extreme values matter more.

Larger outcomes become less likely

$$f(x) \longrightarrow 0.$$

The densities we consider get smaller in the tails.

A battle of **two forces**.

## Infinite versus undefined

For  $Y = g(X)$ , the expectation is **finite** when

$$\mathbb{E}[|Y|] < \infty.$$

**Infinite**

$$\mathbb{E}[Y] = +\infty.$$

The positive contribution is infinite; the negative contribution is finite.

Reversing the signs gives an expectation of  $-\infty$ .

**Undefined**

$$+\infty - \infty.$$

Both contributions have infinite size. We cannot subtract them.

## Cauchy: a ratio of normals

For independent standard normals  $Z_1$  and  $Z_2$ ,

$$X = \frac{Z_1}{Z_2} \sim \text{Cauchy}(0, 1).$$

Its density

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}.$$

This is also the  $t_1$  distribution.

- A denominator **close to zero** can make the ratio huge.
- Even normal ingredients can produce **heavy tails**.

## Cauchy versus normal tails

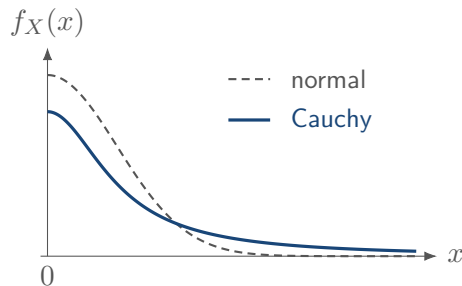
Far into either tail,

$$f_X(x) \approx \frac{1}{\pi x^2}.$$

The Cauchy density shrinks much more slowly than the normal density.

Weighting by  $x$  leaves a  $1/x$  tail:

$$\int_1^M x f_X(x) dx \approx \frac{1}{\pi} \int_1^M \frac{1}{x} dx = \frac{\log M}{\pi} \longrightarrow \infty \quad \text{as } M \rightarrow \infty.$$



**Finite probability** in the tail, but **infinite** first-moment contribution.

## Cauchy moments

For  $X \sim \text{Cauchy}(0, 1)$ :

The mean does not exist

$$\int_{-\infty}^{\infty} x f_X(x) dx \rightsquigarrow \infty - \infty.$$

The two tails contribute opposite signs,  
each with infinite size.

The second moment is infinite

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = +\infty.$$

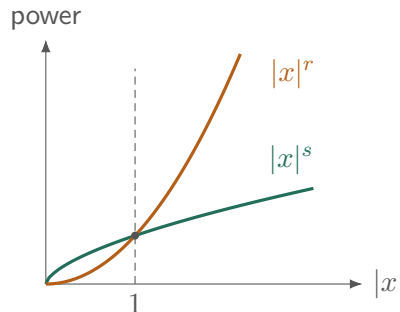
All contributions are nonnegative: there  
is no subtraction to resolve.

The fact that a Cauchy is symmetric doesn't matter!

## From higher to lower moments

For  $0 < s < r$ :

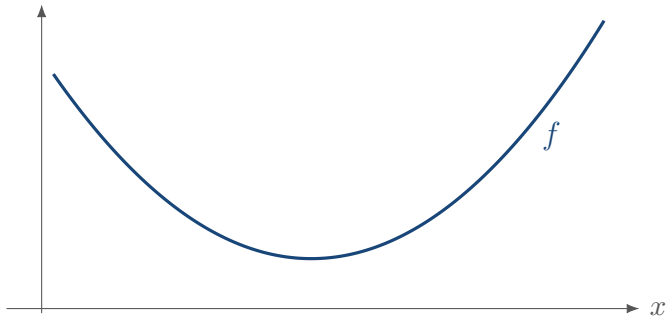
- Higher powers put more weight on extreme outcomes.
- If their average is finite, a lower power cannot make the tails worse.
- Near zero, both powers are bounded: the problem is in the tails.



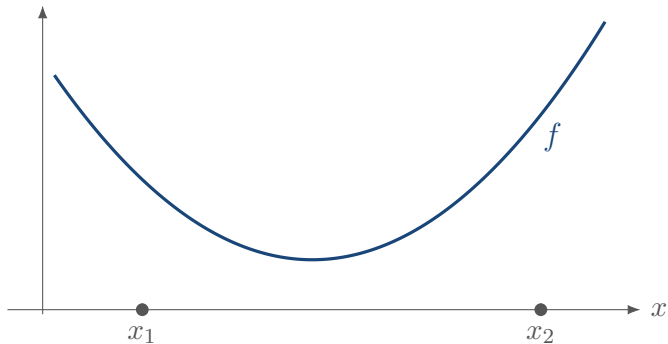
We will use [Jensen's inequality](#) to prove this.

# Jensen's Inequality

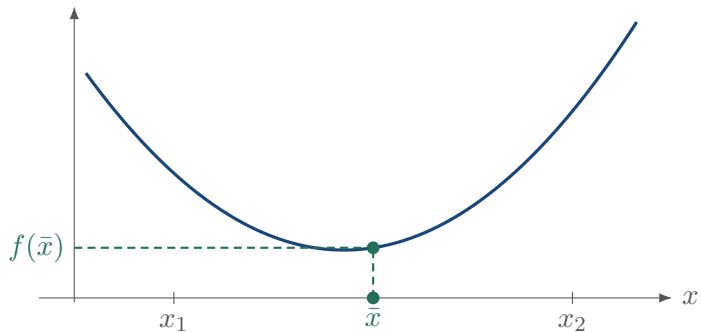
# Jensen's inequality



## Jensen's inequality

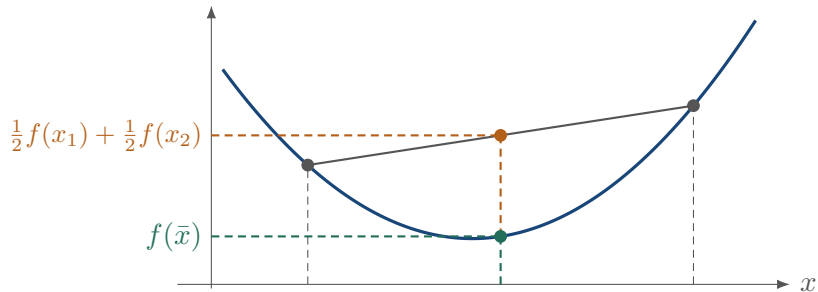


## Jensen's inequality

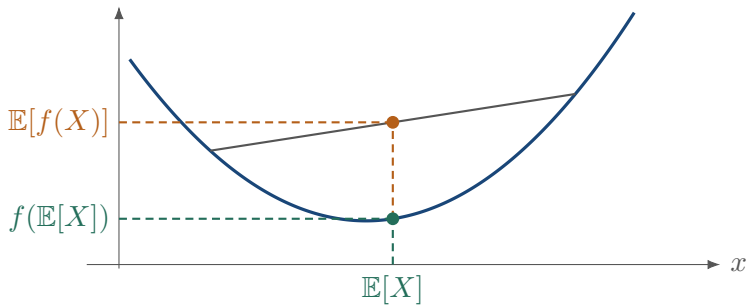


$$\bar{x} = \frac{1}{2}x_1 + \frac{1}{2}x_2.$$

## Jensen's inequality

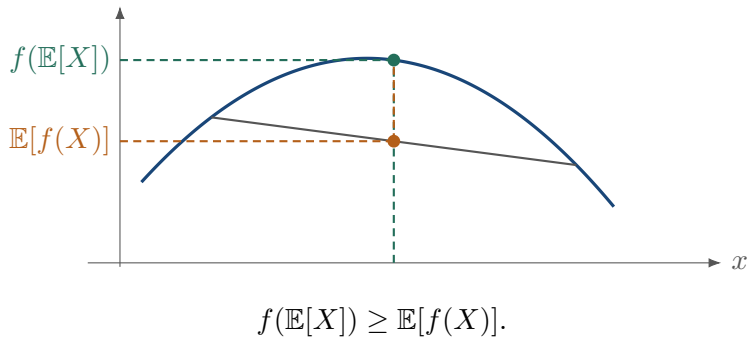


## Jensen's inequality



$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

## Jensen's inequality



## Jensen's inequality: three cases

$$\text{Convex } f : f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)],$$

$$\text{Concave } f : f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)],$$

$$\text{Linear } f : f(\mathbb{E}[X]) = \mathbb{E}[f(X)].$$

For  $f(x) = x^2$ ,

$$(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2].$$

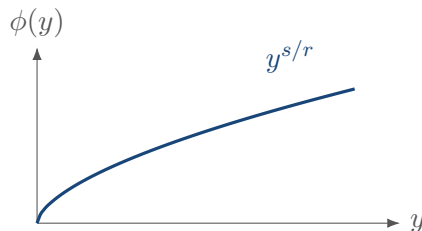
Thus

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \geq 0.$$

## Moment existence via Jensen

Let  $0 < s < r$  and suppose  $\mathbb{E}[|X|^r] < \infty$ .

- Set  $Y = |X|^r$ , so  $Y \geq 0$ .
- Define  $\phi(y) = y^{s/r}$ .
- Since  $0 < s/r < 1$ ,  $\phi$  is concave.



By concave Jensen,

$$\mathbb{E}[|X|^s] = \mathbb{E}[\phi(Y)] \leq \phi(\mathbb{E}[Y]) = (\mathbb{E}[|X|^r])^{s/r} < \infty.$$

# The DGP and OLS

## Three views of a DGP

Scalar:  $y_i = \beta_0 + \beta_1 z_i + \varepsilon_i.$

Vector:  $y_i = x_i' \beta + \varepsilon_i.$

Matrix:  $y = X\beta + \varepsilon.$

## Scalar and vector notation

Start from the simple regression DGP

$$y_i = \beta_0 + \beta_1 z_i + \varepsilon_i.$$

Define

$$x_i = \begin{pmatrix} 1 \\ z_i \end{pmatrix}, \quad x'_i = \begin{pmatrix} 1 & z_i \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

Then

$$x'_i \beta = \begin{pmatrix} 1 & z_i \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \beta_0 + \beta_1 z_i.$$

and so

$$y_i = x'_i \beta + \varepsilon_i$$

## Stacking observations

For each observation  $i = 1, \dots, n$ ,

$$y_i = \beta_0 + \beta_1 z_i + \varepsilon_i.$$

Stack the  $n$  equations into **one matrix equation**:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & z_1 \\ 1 & z_2 \\ \vdots & \vdots \\ 1 & z_n \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

$y \qquad = \qquad X \qquad \beta \qquad + \qquad \varepsilon$

## Checking dimensions

$$\begin{array}{ccccccc} y & = & X & \beta & + & \varepsilon \\ n \times 1 & & n \times k & k \times 1 & & n \times 1 \end{array}$$

$n$ : number of observations.

$k$ : number of regressors, including the intercept.

Here  $k = 2$ : row  $i$  of  $X$  is  $(1, z_i)$ .

$$X = \begin{pmatrix} 1 & z_1 \\ 1 & z_2 \\ \vdots & \vdots \\ 1 & z_n \end{pmatrix}.$$

## $X'X$ : intercept and slope

Each entry is a **sum of products** of two columns of  $X$ .

$$X'X = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ z_1 & z_2 & \cdots & z_n \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix} = \begin{pmatrix} n & \sum_i z_i \\ \sum_i z_i & \sum_i z_i^2 \end{pmatrix}.$$

Intercept with itself

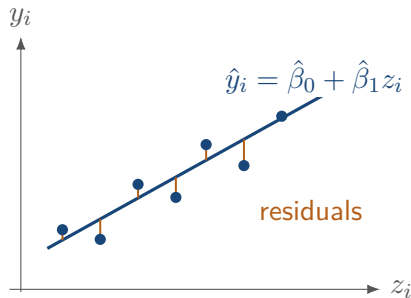
$$\sum_i 1 \cdot 1 = n$$

Intercept with regressor

$$\sum_i 1 \cdot z_i = \sum_i z_i$$

Regressor with itself

$$\sum_i z_i \cdot z_i = \sum_i z_i^2$$



- In the one-regressor case, OLS chooses the best-fitting line:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 z_i.$$

- The residual is the vertical gap:

$$\hat{u}_i = y_i - \hat{y}_i.$$

- OLS minimises the sum of squared residuals:

$$\sum_{i=1}^n \hat{u}_i^2.$$

- With a binary regressor, the OLS slope equals the difference in sample means.

## OLS: with and without an intercept

With  $X$  of full column rank,  $\hat{\beta} = (X'X)^{-1}X'y$ .

The derivation is in the [linear algebra guide](#) and the [appendix](#).

### With an intercept

$$y_i = \beta_0 + \beta_1 z_i + \varepsilon_i$$

$$\hat{\beta}_1 = \frac{\text{sample covariance of } z \text{ and } y}{\text{sample variance of } z}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{z}$$

### Without an intercept

$$y_i = \beta_1 z_i + \varepsilon_i$$

$$\hat{\beta}_1 = \frac{\sum_i z_i y_i}{\sum_i z_i^2}$$

The fitted line passes through zero.

## From assumptions to properties

Recall Lecture 1:

DGP assumptions  $\implies$  unbiasedness, consistency, or a variance formula.

For OLS, the key is **how errors relate to regressors**.

- **Scalar**: describe one observation.
- **Vector**: prove consistency.
- **Matrix**: prove unbiasedness and derive the variance.

Same DGP, different notation for different proofs.

## OLS assumptions

In  $y = X\beta + \varepsilon$ , assumptions concern **population errors**, not residuals.

### Mean

- Unbiasedness:  $\mathbb{E}[\varepsilon \mid X] = 0$ .
- Consistency:  $\mathbb{E}[x_i \varepsilon_i] = 0$ , with an LLN.

### Variance and dependence

$$\Omega = \text{Var}(\varepsilon \mid X) = \mathbb{E}[\varepsilon \varepsilon' \mid X] \quad \text{when } \mathbb{E}[\varepsilon \mid X] = 0.$$

We also need rank, sampling and moment conditions.

## Covariance assumptions: $3 \times 3$

Homoscedastic, no autocorrelation

$$\Omega = \begin{pmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{pmatrix}$$

For  $n$  observations:  $\Omega = \sigma^2 I_n$ .

Heteroscedastic, no autocorrelation

$$\Omega = \begin{pmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{pmatrix}$$

## General error covariance

Serial correlation / dependence

$$\Omega = \begin{pmatrix} \sigma_1^2 & \omega_{12} & \omega_{13} \\ \omega_{12} & \sigma_2^2 & \omega_{23} \\ \omega_{13} & \omega_{23} & \sigma_3^2 \end{pmatrix}$$

Clustered example

$$\Omega = \begin{pmatrix} \sigma_1^2 & \omega_{12} & 0 \\ \omega_{12} & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{pmatrix}$$

## The variance of OLS

With  $X$  of full column rank, write

$$\hat{\beta} = \beta + A\varepsilon, \quad A = (X'X)^{-1}X'.$$

Conditional on  $X$ ,  $A$  is fixed. With  $\Omega = \text{Var}(\varepsilon \mid X)$ ,

$$\text{Var}(\hat{\beta} \mid X) = A\Omega A' = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

Homoscedastic, uncorrelated errors:  $\Omega = \sigma^2 I_n$ .

$$\text{Var}(\hat{\beta} \mid X) = \sigma^2 (X'X)^{-1} \underbrace{X'X(X'X)^{-1}}_I = \sigma^2 (X'X)^{-1}.$$

## Key messages

- Order statistics describe sorted observations, including extremes.
- Combining normals gives us  $\chi^2$ ,  $t$  and  $F$  distributions.
- Tail behaviour determines whether moments are finite.
- Jensen's inequality links curvature and expectations.
- Matrix notation collects regression equations and covariances.
- Standard errors must reflect the error structure.

Next time

# Lecture 4: Hypothesis Testing

## OLS: expand the objective non-examinable

OLS chooses  $b$  to minimise  $SSR(b) = \sum_{i=1}^n (y_i - x_i' b)^2$ .

$$\begin{aligned} SSR(b) &= (y - Xb)'(y - Xb) \\ &= y'y - y'Xb - b'X'y + b'X'Xb. \end{aligned}$$

The middle two terms are scalars, so  $y'Xb = b'X'y$ . Hence

$$SSR(b) = y'y - 2b'X'y + b'X'Xb.$$

- $y'y$  does not depend on  $b$ .
- $-2b'X'y$  is the linear term.
- $b'X'Xb$  is the quadratic term.

Using multivariate calculus in matrix notation,

$$\nabla_b SSR(b) = -2X'y + 2X'Xb.$$

At the minimum, set the gradient equal to zero:

$$-2X'y + 2X'X\hat{\beta} = 0.$$

Therefore

$$X'X\hat{\beta} = X'y.$$

If  $X'X$  is invertible,

$$\hat{\beta} = (X'X)^{-1}X'y.$$

We still need to check that this stationary point is the unique minimum.

## OLS: verify the minimum non-examinable

Assume  $X$  has full column rank. For any  $h \neq 0$ ,

$$h'X'Xh = \|Xh\|^2 > 0,$$

so  $X'X$  is positive definite and invertible.

Let  $\hat{u} = y - X\hat{\beta}$ . The normal equations give  $X'\hat{u} = 0$ . For any alternative  $b = \hat{\beta} + h$ ,

$$\begin{aligned} SSR(\hat{\beta} + h) &= (\hat{u} - Xh)'(\hat{u} - Xh) \\ &= SSR(\hat{\beta}) - 2h'X'\hat{u} + h'X'Xh \\ &= SSR(\hat{\beta}) + \|Xh\|^2. \end{aligned}$$

Every  $h \neq 0$  raises the objective:  $\hat{\beta} = (X'X)^{-1}X'y$  is the **unique global minimum**.